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Time taken	7 secs
Marks	0/6
Grade	0 out of 100

Question 1

Not answered

Marked out of 1

Which of the following statements is true concerning the probability of an event occurring?

Select one:

- ☐ An event whose probability is a negative value, will occur very rarely
- ☐ The sum of all individual probabilities of different outcomes must be equal to the number of outcomes
- ☐ An event with probability of zero can occur but at a low frequency
- ☐ An event with probability of 1 will always occur
- ☐ The probability of any event must be between -1 and +1

Your answer is incorrect.

The standard equation for a probability calculation is $P(\text{event}) = A/N$, where $P(\text{event})$ is the probability of the studied event occurring, A is the number of times the desired event actually occurs, and N is the total number of trials during which A may occur. The probability of A occurring is expressed as a percentage that is never greater than 1 (100%) or less than 0 (0%).

The correct answer is: An event with probability of 1 will always occur

Question **2**

Not answered

Marked out of 1

You are waiting for your next new patient in the OPD. Your clinical team books only three types of new patients for you to see - borderline personality, schizophrenia or depression. Concerning the probability of your next patient having a diagnosis of borderline personality, which of the following is true?

Select one:

- ☐ The probability is $1/3$
- ☐ The probability is $1/2$
- ☐ The probability is zero
- ☐ The probability is 1
- ☐ The probability is 3

Your answer is incorrect.

Out of 3 different types of events, the probability of one event to occur is $1/3$.

The correct answer is: The probability is $1/3$

Question 3

Not answered

Marked out of 1

You are waiting for your next new patient in the OPD. Your clinical team books only three types of new patients for you to see - borderline personality, schizophrenia or depression. You are aware that a high degree of comorbidity among the three disorders is noted in your clinical sample. Which of the following can be said about the probability of your next patient having one of the three diagnoses, assuming random booking?

Select one:

- ☐ If the next patient has borderline disorder, she cannot have schizophrenia.
- ☐ The chance that the next patient has any of the three diagnoses is 3
- ☐ The chance that the next patient is neither depressed nor having borderline personality is around 2/3
- ☐ The chance that the next patient has none of the three diagnoses is one.
- ☐ The three events are not mutually exclusive

Your answer is incorrect.

Due to a high degree of co-morbidity, the events become dependent on each other i.e. they are not mutually exclusive. The probability that the next patient has any of the three must be less than one - it cannot be 3. The chance that they have neither of the three is zero - it cannot be one given the fact that your team books only one of these three patients for you to see. The chance that the next patient is neither depressed nor having borderline personality must be around 1/3.

The correct answer is: The three events are not mutually exclusive

Question 4

Not answered

Marked out of 1

The chance of having Down syndrome if born to a mother aged 35 is 2%. The chance of having Williams Syndrome if born to a mother aged 35 is 0.5%. What is the probability of being born with either Down or Williams syndrome if the mother is aged 35, assuming the two syndromes occur independent of each other and cases with both Down and Williams syndrome have been reported?

Select one:

- ☐ 0.25
- ☐ 0.015
- ☐ 0.1
- ☐ 0.01
- ☐ 0.024

Your answer is incorrect.

For two events, the probability that either will occur is the sum of their probabilities, minus the probability that both will occur (if they are not mutually exclusive i.e. they are not like toss of coin where having a head means not having a tail or vice versa). This is because a simple addition of two probabilities will result in the shared probability (Down AND Williams) getting counted twice. Hence the combined probability is subtracted from the sum of the probabilities: $p(\text{Williams or Down}) = p(\text{Williams}) + p(\text{Down}) - p(\text{Down AND Williams})$. For events that are independent of each other, as stated here, the probability that both will occur is the product of their probabilities. So $p(\text{Down AND Williams}) = 2\% \times 0.5\% = 0.01\%$ [note that the denominator had two 100s]. Then $p(\text{Down OR Williams}) = 2\% + 0.5\% - 0.01\% = 2.49\%$. If two conditions are mutually exclusive such as coin toss, then the probability that either one will occur is simply the sum of their probabilities. The closest answer is 0.024.

The correct answer is: 0.024

Question 5

Not answered

Marked out of 1

In an acute admission ward, 25 patients are admitted. There are ten patients with psychosis and 15 with neurosis. If one randomly selects a patient the odds of him/her being psychotic is

Select one:

- ☐ (10/25)
- ☐ (10/15)
- ☐ 150/25
- ☐ Not enough information to calculate the odds
- ☐ (15/25)

Your answer is incorrect.

Odds = frequency of one event occurring / frequency of another event. Probability = frequency of one event occurring / frequency of all events of interest. Here odds of being psychotic are = 10/15. If probability is asked then the probability of being psychotic = $10/25 = 2/5$

The correct answer is: (10/15)

Question 6

Not answered

Marked out of 1

In an acute admission ward, 25 patients are admitted. If the odds of a randomly selected patient being psychotic are $2/3$, then the probability of finding a psychotic patient from the sample is

Select one:

- ☐ (2/5)
- ☐ (1/5)
- ☐ (3/5)
- ☐ (1/3)
- ☐ Not enough information to calculate the probability

Your answer is incorrect.

Odds = frequency of one event occurring / frequency of other event
Probability = frequency of one event occurring / frequency of all events of interest
The relationship between odds (O) and probability (P) is $P = O / (1+O) = (2/3) / (1 + 2/3) = 2/3 / (5/3) = 2/5$

The correct answer is: (2/5)



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